Michael Boiman

Quality Engineering & AI Automation Expert • 4+ Years AI Development • 20+ Years Quality Assurance

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-80%
weniger Rückfragen
durch Quality-Monitoring

96%
Qualität nach Migration
vorher 30%

80%
KI-gestützte Tests
auto-generiert

Quality Engineering & AI Competencies

Quality Engineering & AI Automation
Quality-first automation: Email bots, invoice processing, workflow automation with 80% workload reduction, measurable cost savings, processing time from hours to seconds

AI Quality Engineering Excellence
80% auto-generated tests & test cases, IDE integration, Go/No-Go in minutes instead of hours, 50% less development time, 80% fewer inquiries

Next-Gen AI Protocols & Agentic Systems
MCP & A2A protocol implementation, Google Agent Development Kit (ADK), multi-agent orchestration, JSON-RPC, secure agent-to-agent communication, enterprise-ready

RAG Systems & Enterprise NLP
Vector embeddings, LangChain, Q&A bots, spaCy/NLTK, 95% accuracy, EU standards, contextual intelligence

Multi-Cloud AI Architecture
Azure OpenAI, Google Cloud AI, serverless MLOps, auto-scaling, real-time monitoring, production-grade deployment

Transparency & Business Intelligence
Real-time dashboards, quality visualization, end-to-end monitoring, KPI automation, single source of truth

Career Profile

Quality Engineer and AI Automation Expert with 20+ years in software quality assurance and 4 years of specialized AI development. My focus: transforming business processes through measurable automation — from test generation to invoice processing.

In enterprise projects (DB Vertrieb, DVAG, TÜV Süd), I built quality monitoring systems that cut Go/No-Go decisions from hours to minutes, introduced AI-driven test automation that generates 80% of test cases automatically, and validated legacy migrations 24/7 — improving quality from 30% to 96%.

Today I combine quality engineering with AI architecture: MCP and A2A protocols, multi-agent systems, e-invoicing platforms to EU standards, and a 40+ skill automation ecosystem for enterprise workflows.

TECHNICAL SKILLS

Quality Engineering | Playwright | Cucumber/Gauge | Python | LLMs (Claude/Gemini/GPT/Copilot) | MCP & A2A Protocols | Google ADK | LangChain | Peppol/E-Invoicing | CI/CD | Kubernetes | Azure Functions | Grafana | Elasticsearch | REST APIs | Docker | JMeter | Gatling

EDUCATION

Diploma in Computer Science (UAS)
Cologne University of Applied Sciences, Gummersbach
2000 – 2005

ISTQB Certified Tester
2005

Certified Dynatrace Diagnostics Basic Training
2025

LANGUAGES

German	Native
English	Professional
Russian	Conversational
Hebrew	Conversational

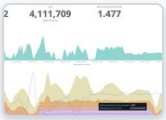
ADDITIONAL QUALIFICATIONS

ISTQB Certified | Agile (Scrum/Kanban)

4+ Years AI Development, 20+ Years Quality Engineering

PROJECTS & SOLUTIONS

24/7 Automated Legacy Migration Validator: Dual-Run Quality Engineering



Java | Cucumber | Kibana | Graylog | Elasticsearch | Container | CI/CD | Mainframe | Dashboard Engineering | Test Automation

Enterprise Legacy Transformation with Fully Automated Quality Assurance The Challenge: Complex mainframe price calculation with millions of combination possibilities needed secure migration - without quality loss or data errors. **The...

Quality Dashboard - Real-time Overview

Grafana | Elasticsearch | Python | CI/CD | Playwright | Jira API | Confluence API | LeanIX | PDF Generation

Fully automated end-to-end solution for real-time quality monitoring across 10 test tools, 5 environments plus multi-source integration (Pipeline, LeanIX, Jira, Confluence) for automated PDF reports with open issues, bugs, and...

E-Invoicing Automation Platform

Python | Azure Functions | Peppol | Storecove | SAP ByDesign | ZUGFeRD | Factur-X | UBL | Hexagonal Architecture

Fully Automated E-Invoice Processing to EU Standards Core Features: • **Peppol Network Integration**: Connection to European e-invoicing network via Storecove Access Point • **Multi-Format Support**: ZUGFeRD, Factur-X, XRechnung, UBL 2.1,...

Email Classification and Processing Workflow

Python | Azure Functions | OpenAI GPT-3.5 | Microsoft Graph API | Elasticsearch | Kibana | Docker

Complete automation solution with Microsoft Graph API, SAP RFC, Azure Functions (Python), OpenAI GPT-3.5-Turbo, Elasticsearch 8.x, Kibana Dashboards, Docker containers, and GitHub Actions CI/CD - 80% reduction in manual work.

AI Context Orchestrator — Multi-Repo Development Platform

Python | YAML | Claude Code Plugins | Hooks API | GitHub Actions | Elasticsearch | Azure Functions | Microsoft Graph API

Central Control System for Enterprise AI Development Core Features: • **Context Orchestrator**: ecosystem.yaml as single source of truth for 20+ repos, customers, infrastructure and skills • **40+ Production Skills**: Log...

AI-Driven Chatbot for Trade-Fair Interaction

Python | OpenAI GPT-4 | Streamlit | FastAPI | Vector Embeddings

Development of an interactive chatbot using RAG (Retrieval-Augmented Generation), OpenAI GPT-4, Vector Embeddings, Python FastAPI, and Streamlit frontend to optimize user interaction on trade-fair websites.

Enterprise AI Consulting: From Strategy to Implementation

AI Strategy | Process Analysis | ROI Assessment | Implementation Planning | Change Management | Executive Workshops | Quality Engineering

Target Audience: C-Level, Management, Product Owners, Business Analysts **Consulting Services:** • **Strategy Development:** From AI vision to measurable implementation roadmap • **Process Analysis:** Identification of automation potential, quality-first assessment •...

Developer Workshop: AI-Agent Development & Integration

Anthropic Claude | Google Gemini | OpenAI GPT-4o | MCP | A2A | Playwright | JSON-RPC | IDE Integration | CI/CD

Target Audience: Developers, Technical Leads, DevOps Engineers **Workshop Content:** • **Hands-on AI Agent Development:** Live coding with Claude, Gemini, GPT-4o • **Next-Gen Protocols:** MCP & A2A implementation for enterprise integration...

Medical Transcription System

Python | Whisper | Pyannote | Ollama | Llama3 | Apple Silicon | SOAP Protocol

GDPR-Compliant Transcription of Doctor-Patient Conversations Core Features: • **100% Local**: No cloud uploads, all processing on local machine (Privacy-by-Design) • **Speaker Diarization**: Automatic doctor/patient distinction with Pyannote 3.1 (2.6% WER)...

AI Learning Platform

Astro | React 19 | TypeScript | Supabase | EU AI Act

EU AI Act Compliant Learning Platform for AI Workshops Core Features: • **Article 4 Compliance**: EU AI Act compliant certifications with audit trail • **15+ Learning Card Types**: Interactive quizzes,...

PROFESSIONAL EXPERIENCE

LLM Infrastructure Architect & Automation Engineer

06/2025 – present

BKS - AI Research & Development

Development of LLM-orchestrated Enterprise Development Ecosystem: Fully integrated system for knowledge management, project automation, and development workflows with equal expertise in Claude, Gemini, and OpenAI GPT.

Tools & Technologies LLM Orchestration: Claude, Google Gemini, OpenAI GPT, GitHub Copilot (equal expertise), Model Context Protocol (MCP), Google Agent Development Kit (ADK) · AI Development Platform: Claude Code, Plugin Architecture (Skills, Hooks, Commands, Agents), YAML/Markdown Configuration · Multi-Agent: A2A...

Quality Engineer

09/2025 – 02/2026

TÜV Süd

Quality Engineering for an enterprise platform for AI-powered document processing in medical device certification (MDR/IVDR) with AI chat integration and data pipeline automation.

Tools & Technologies Testing: Behave, Playwright, TypeScript, Python, pytest, Page Object Model, Fixtures · Monitoring: Azure Monitor Workbooks, KQL, Application Insights, Log Analytics · Backend: FastAPI, Python, Pydantic, SQLModel, Azure Functions, Azure Service Bus · Frontend: React, TypeScript, Vite, TanStack...

AI-driven Automated QA Environment for Energy Infrastructure

07/2024 – 01/2025

AkkuSwap Startup

QA Leadership for EU-wide Battery Swap Infrastructure: AI-driven simulation tool for battery swap station network with energy infrastructure integration

Technical Stack Infrastructure: Linux, Docker, Azure OpenAI · Automation: pytest, AI code generation integration patterns · Monitoring: Grafana, Azure Monitoring · Energy Systems: Battery swap infrastructure, grid integration simulation · Established QA methodology for AI-driven infrastructure simulation · Created testing...

Platform Quality Architect (CI/CD, Monitoring, Automation)

08/2021 – 05/2025

DVAG

Test management & coordination, as well as introduction of CI/CD with quality-gate setup. Acting as cross-team Quality Lead: onboarding, guidance, consulting and training for the introduction of quality metrics and test automation within the pipeline across teams. **Key responsibilities** - Design and implementation of functional and technical dashboards with Grafana...

Tools IntelliJ / Visual Studio Code · Java / JavaScript / TypeScript · Playwright / Gauge / Karate · Python for processing and analysis of test and reporting data · CI/CD / GitHub – reusable workflows · Helm / Containers...

Technical Lead & AI Automation Architect

01/2024 – 04/2025

BKS

Technical lead & development: Design and implementation of a scalable automation solution for processing e-invoices and PDF invoices fetched from a mailbox of a leading online job platform. Full responsibility from requirements analysis through implementation to production go-live.

Tools & technologies Azure Functions: Serverless orchestration of automation workflows. · Python: Core logic for data extraction, classification and integration with automated CI/CD processes, packaging, and comprehensive unit and integration tests for quality assurance. · E-Invoicing: Peppol Network, Storecove API,...

FRÜHERE POSITIONEN

AI Engineering Lead & ML Solutions Architect — BKS on behalf of Ryze	04/2023 - 04/2024
DevOps Quality Lead & Dashboard Architect — DB Vertrieb	01/2017 - 05/2021
Performance Engineering Lead & Test Architect — DB Systel	09/2015 - 01/2017
Senior iOS Developer & Mobile Architect — Telekom	04/2015 - 09/2015
Quality Assurance Lead & Test Automation Architect — Siemens	05/2009 - 03/2015
Performance Test Engineer & Load Testing Specialist — ING-DIBA	03/2009 - 05/2009
Performance Test Engineer & Quality Consultant — British Telecom, Mobiliar, DB-Systel, Sparkassen Informatik, Loyalty Partner, Telekom, Itelium, Deutsche Post, Postbank	01/2006 - 12/2008
Workshop: Development with Generative Language Models — Developer Workshop - Agent-based Software Development	06/2025
Presentation: Efficient Documentation through Automation — Enterprise Presentation - AI-powered Documentation Workflows	04/2025
AI Workshop for Business Integration — Applications of AI in Business Context	2023